

BSC106/FSC105

General Physics

Mechanics 1

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1.3 Kinematics in 2D and 3D

1.3.1 Vectors in two dimensions

- To describe the motion of a particle in space, we must first be able to describe the particle's position
- In 2D Cartesian plane, we use vectors in $\mathbb{R}^2$

1. **Components:** A vector r can be given by its components $\vec{r} = x\hat{i} + y\hat{j}$ ✓

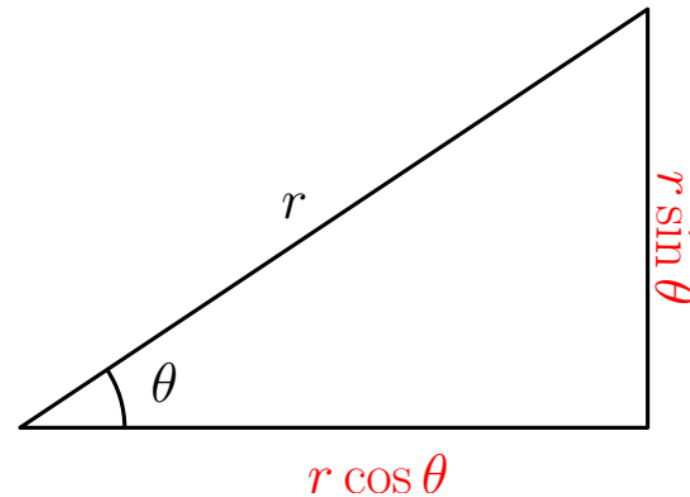
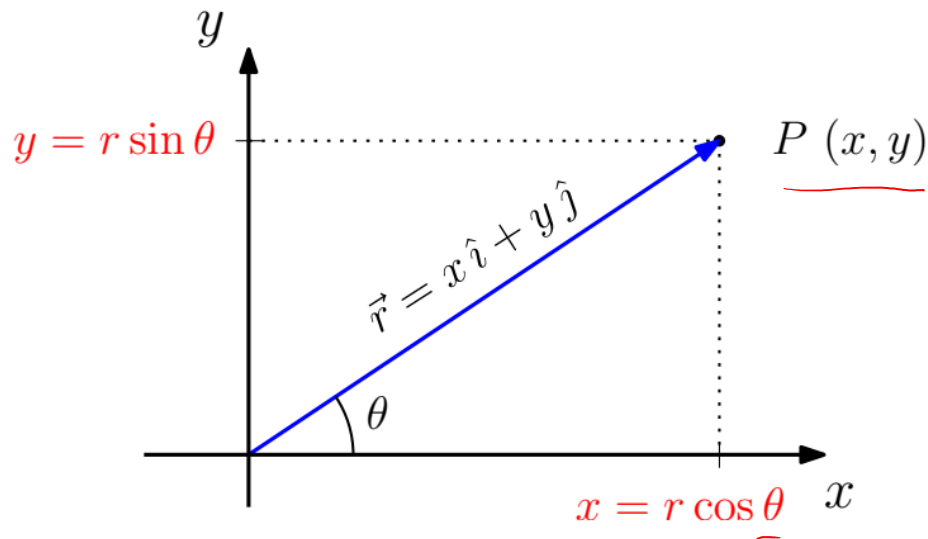
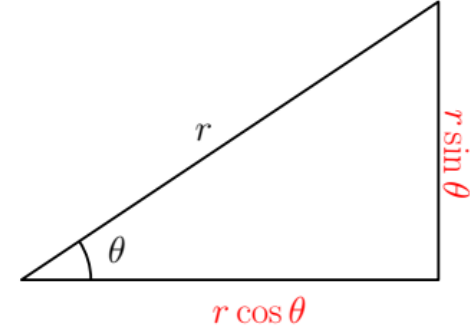
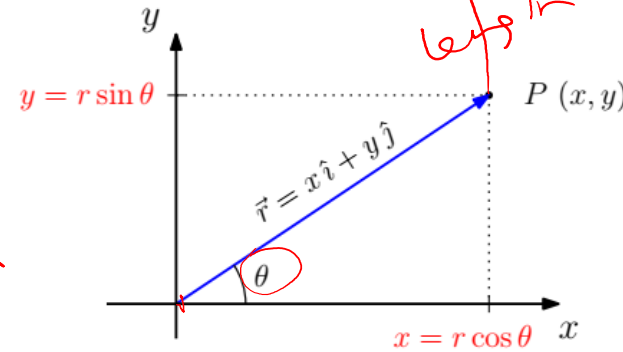


Figure 1.4: Vectors in $\mathbb{R}^2$.

2. **Magnitude** and **direction**: A vector can be specified by its **magnitude** (also called 'length' or 'norm') and **direction**. The length of a vector is given by $r = \sqrt{x^2 + y^2}$ and its direction can be given by its angle with the x -axis can be calculated by $\tan \theta = \frac{y}{x}$ (see Fig. 1.4.)

Therefore we may write a vector as

$$\begin{aligned}
 \vec{r} &= x \hat{i} + y \hat{j} \\
 &= (r \cos \theta) \hat{i} + (r \sin \theta) \hat{j} \\
 &= r (\underbrace{\cos \theta \hat{i} + \sin \theta \hat{j}}_{=\hat{r}}) \\
 &= r \hat{r}.
 \end{aligned}
 \tag{1.34}$$



Here, we have defined $\hat{r} = \cos \theta \hat{i} + \sin \theta \hat{j}$ to be the **unit vector**. It carries the information about the direction of the vector. It is a unit vector because its length is 1,

$$|\hat{r}| = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1. \tag{1.35}$$

where $\hat{i}$ and $\hat{j}$ are **unit vectors** in the x - and y -directions, respectively.

3. Addition of vectors: Two vectors

$$\vec{A} = a_x \hat{i} + a_y \hat{j}, \quad \vec{B} = b_x \hat{i} + b_y \hat{j},$$

can be added together as follows:

$$\begin{aligned} \vec{A} + \vec{B} &= a_x \hat{i} + a_y \hat{j} + b_x \hat{i} + b_y \hat{j} \\ &= (a_x + b_x) \hat{i} + (a_y + b_y) \hat{j}. \end{aligned}$$

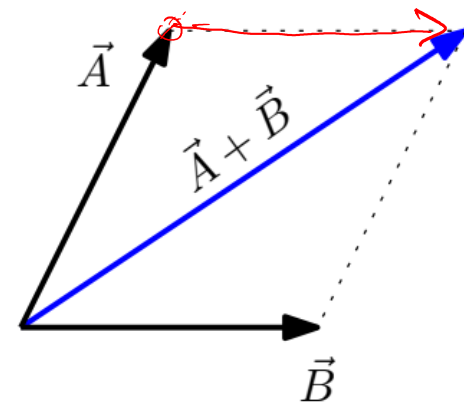


Figure 1.5: Vector addition

4. **Subtraction of vectors:** The subtraction of B from A can be done as follows:

$$\begin{aligned}\vec{A} - \vec{B} &= \vec{A} + (-\vec{B}) \\ &= a_x \hat{i} + a_y \hat{j} - b_x \hat{i} - b_y \hat{j} \\ &= (a_x - b_x) \hat{i} + (a_y - b_y) \hat{j}.\end{aligned}\tag{1.38}$$

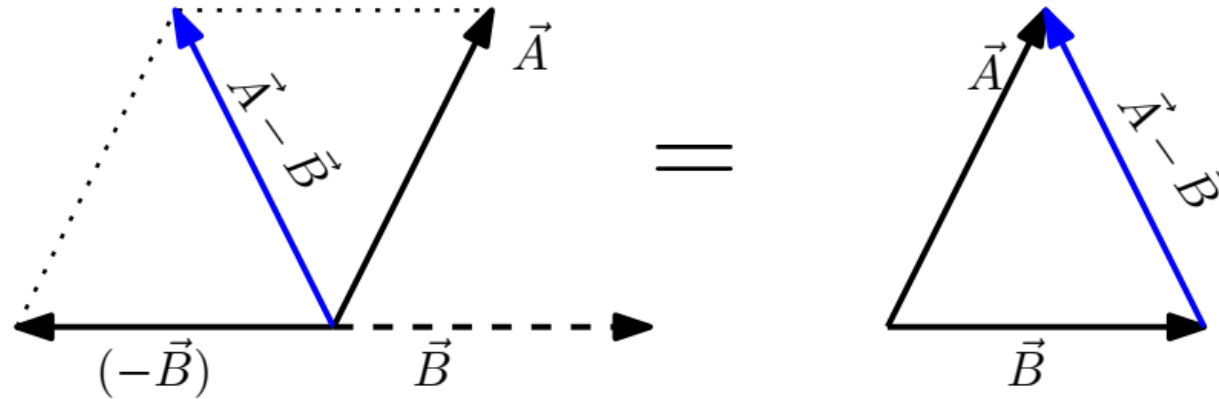


Figure 1.6: Vector subtraction

5. **Dot product:** The dot product between two vectors is given by

$$\vec{A} \cdot \vec{B} = AB \cos \alpha, \quad (1.39)$$

where A and B are the lengths of $\vec{A}$ and $\vec{B}$ respectively, and α is the angle between the two vectors (drawn in such a way that their tails are joined):

In component form, we use the rule

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = 1, \quad \hat{i} \cdot \hat{j} = 0.$$

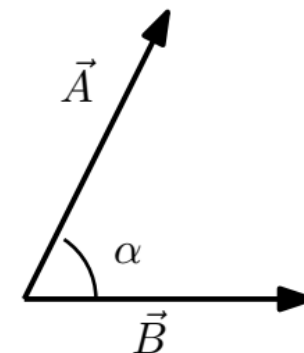


Figure 1.7: Dot product.

Then the dot product can also be written as

$$\begin{aligned} \vec{A} \cdot \vec{B} &= (a_x \hat{i} + a_y \hat{j}) \cdot (b_x \hat{i} + b_y \hat{j}) \\ &= a_x b_x \hat{i} \cdot \hat{i} + a_y b_y \hat{j} \cdot \hat{j} + (a_x b_y + a_y b_x) \hat{i} \cdot \hat{j} \\ &= a_x b_x + a_y b_y, \end{aligned}$$

There are some special cases where

- $\vec{A} \cdot \vec{A} = a_x^2 + a_y^2 = |\vec{A}|^2.$
- $\vec{A} \cdot \hat{i} = a_x, \quad \vec{A} \cdot \hat{j} = a_y.$
- $\sqrt{\hat{i} \cdot \hat{i}} = |\hat{i}| = 1$ and $\sqrt{\hat{j} \cdot \hat{j}} = |\hat{j}| = 1$

1.3.2 Vectors in three dimensions

Vectors in $\mathbb{R}^3$

1. **Components, magnitude, and direction:** A vector in $\mathbb{R}^3$ can be given in Cartesian coordinates as

$$\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}.$$

$$r = |\vec{r}| = \sqrt{x^2 + y^2 + z^2},$$

Alternatively, we can give a vector in spherical coordinates as

$$x = r \cos \phi \sin \theta, \quad y = r \sin \phi \sin \theta, \quad z = r \cos \theta,$$

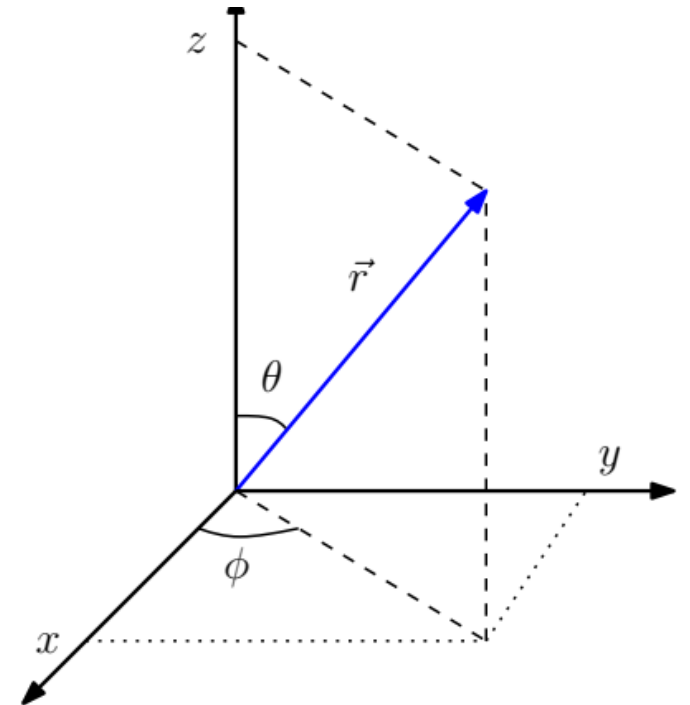


Figure 1.8: Vector in $\mathbb{R}^3$.

2. **Addition and subtraction:** Two vectors

$$\vec{A} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}, \quad \vec{B} = b_x \hat{i} + b_y \hat{j} + b_z \hat{k},$$

$$\vec{A} \pm \vec{B} = (a_x \pm b_x) \hat{i} + (a_y \pm b_y) \hat{j} + (a_z \pm b_z) \hat{k}.$$

3. **Dot product:** The dot product between two vectors is given by

$$\vec{A} \cdot \vec{B} = a_x b_x + a_y b_y + a_z b_z = AB \cos \alpha,$$

where α is the angle between $\vec{A}$ and $\vec{B}$.

4. **Cross product:** The cross product between $\vec{A}$ and $\vec{B}$ produces a new vector $\vec{C}$:

$$\vec{C} = \vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$
$$= (a_y b_z - a_z b_y) \hat{i} - (a_x b_z - a_z b_x) \hat{j} + (a_x b_y - a_y b_x) \hat{k}.$$

The magnitude of $\vec{C}$ is $|\vec{C}| = AB \sin \alpha$,

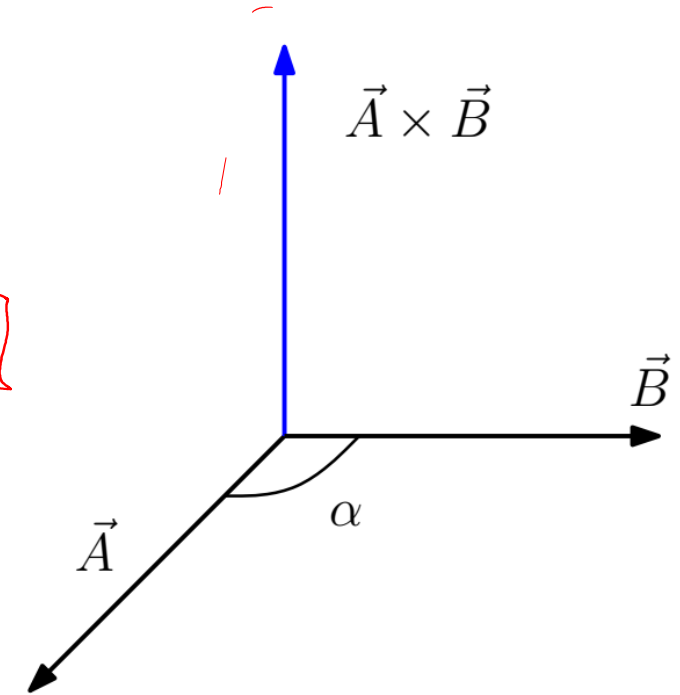


Figure 1.9: The cross product.

where α is the angle between the vectors $\vec{A}$ and $\vec{B}$. The direction is determined by the **right-hand rule**.

In 3 dimension

- Position

$$\vec{r}(t) = x(t) \hat{i} + y(t) \hat{j} + z(t) \hat{k}$$

- Displacement

$$\vec{s} = \Delta \vec{r} = \vec{r}(t_2) - \vec{r}(t_1), \quad t_2 > t_1.$$

- Average velocity

$$\vec{v}_{\text{av}} = \frac{\Delta \vec{r}}{\Delta t} = \frac{\vec{r}(t_2) - \vec{r}(t_1)}{t_2 - t_1}.$$

- Instantaneous velocity $\vec{v} = \lim_{\Delta t \rightarrow 0} \vec{v}_{\text{av}} = \frac{d\vec{r}}{dt}$
$$= \frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} + \frac{dz}{dt} \hat{k}$$
$$= v_x \hat{i} + v_y \hat{j} + v_z \hat{k},$$

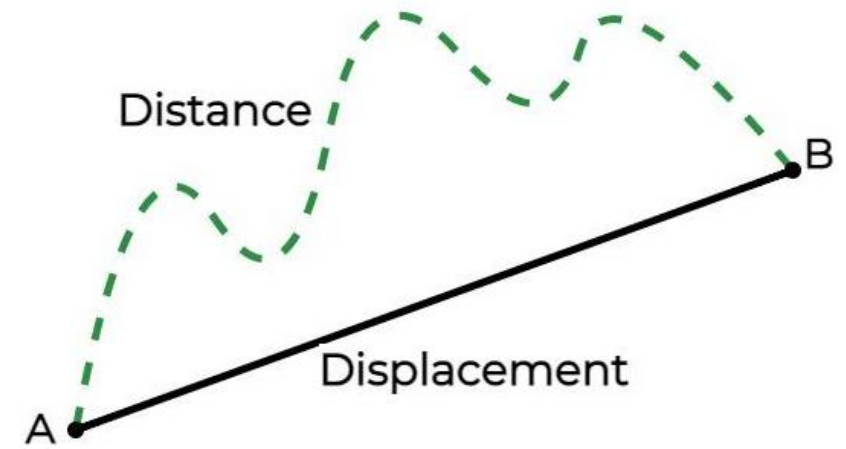
- Average acceleration

$$\vec{a}_{\text{av}} = \frac{\Delta \vec{v}}{\Delta t},$$

- Instantaneous acceleration

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \vec{a}_{\text{av}} = \frac{d\vec{v}}{dt}.$$

Scalar Quantities	Vector Quantities
Have magnitude but no direction	Have magnitude and direction
Distance ✓ Speed Mass Energy Density Power Length, Area, Volume Time Temperature Work	Displacement ✓ Velocity ✓ Weight Acceleration ✓ Force ✓ Impulse ✓ Pressure ✓ Momentum ✓ Gravity Drag ✓



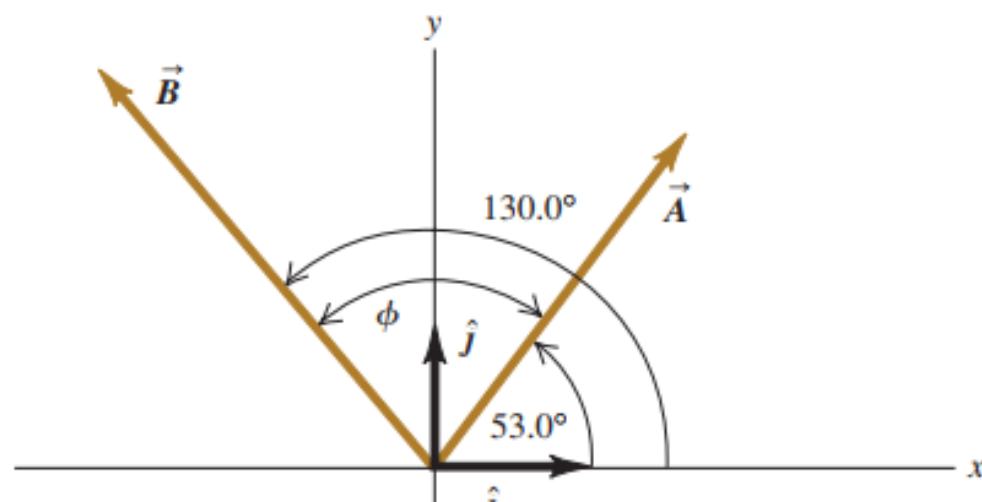
Example 1.10 Calculating a scalar product

Find the scalar product $\vec{A} \cdot \vec{B}$ of the two vectors in Fig. 1.27. The magnitudes of the vectors are $A = 4.00$ and $B = 5.00$.

SOLUTION

IDENTIFY and SET UP: We can calculate the scalar product in two ways: using the magnitudes of the vectors and the angle between them (Eq. 1.18), and using the components of the vectors (Eq. 1.21). We'll do it both ways, and the results will check each other.

1.27 Two vectors in two dimensions.



EXECUTE: The angle between the two vectors is $\phi = 130.0^\circ - 53.0^\circ = 77.0^\circ$, so Eq. (1.18) gives us

$$\vec{A} \cdot \vec{B} = AB \cos \phi = (4.00)(5.00) \cos 77.0^\circ = 4.50$$

To use Eq. (1.21), we must first find the components of the vectors. The angles of $\vec{A}$ and $\vec{B}$ are given with respect to the $+x$ -axis and are measured in the sense from the $+x$ -axis to the $+y$ -axis, so we can use Eqs. (1.6):

$$A_x = (4.00) \cos 53.0^\circ = 2.407$$

$$A_y = (4.00) \sin 53.0^\circ = 3.195$$

$$B_x = (5.00) \cos 130.0^\circ = -3.214$$

$$B_y = (5.00) \sin 130.0^\circ = 3.830$$

As in Example 1.7, we keep an extra significant figure in the components and round at the end. Equation (1.21) now gives us

$$\begin{aligned}\vec{A} \cdot \vec{B} &= A_x B_x + A_y B_y + A_z B_z \\ &= (2.407)(-3.214) + (3.195)(3.830) + (0)(0) = 4.50\end{aligned}$$

EVALUATE: Both methods give the same result, as they should.

Example 1.11 Finding an angle with the scalar product

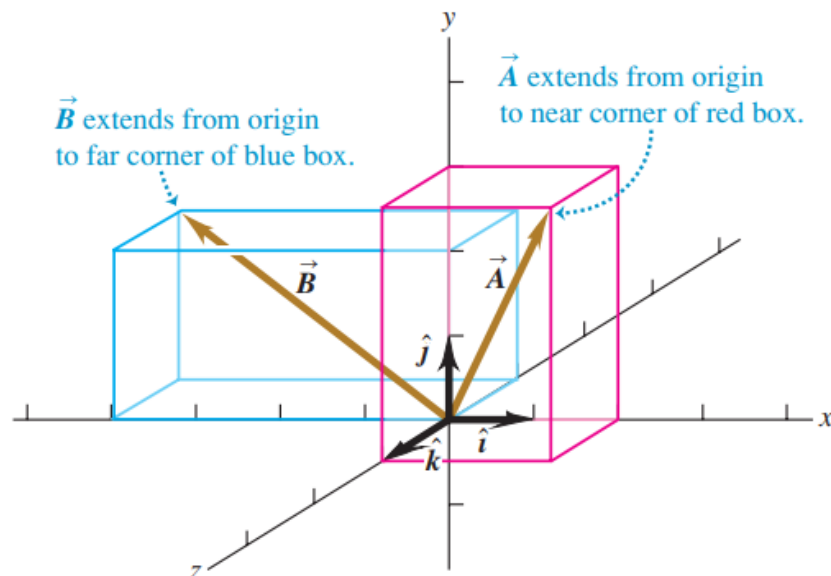
Find the angle between the vectors

$$\begin{aligned}\vec{A} &= 2.00\hat{i} + 3.00\hat{j} + 1.00\hat{k} & \text{and} \\ \vec{B} &= -4.00\hat{i} + 2.00\hat{j} - 1.00\hat{k}\end{aligned}$$

SOLUTION

IDENTIFY and SET UP: We're given the x -, y -, and z -components of two vectors. Our target variable is the angle ϕ between them (Fig. 1.28). To find this, we'll solve Eq. (1.18), $\vec{A} \cdot \vec{B} = AB \cos \phi$, for ϕ in terms of the scalar product $\vec{A} \cdot \vec{B}$ and the magnitudes A and B . We can evaluate the scalar product using Eq. (1.21),

1.28 Two vectors in three dimensions.



$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$, and we can find A and B using Eq. (1.7).

EXECUTE: We solve Eq. (1.18) for $\cos \phi$ and write $\vec{A} \cdot \vec{B}$ using Eq. (1.21). Our result is

$$\cos \phi = \frac{\vec{A} \cdot \vec{B}}{AB} = \frac{A_x B_x + A_y B_y + A_z B_z}{AB}$$

We can use this formula to find the angle between *any* two vectors $\vec{A}$ and $\vec{B}$. Here we have $A_x = 2.00$, $A_y = 3.00$, and $A_z = 1.00$, and $B_x = -4.00$, $B_y = 2.00$, and $B_z = -1.00$. Thus

$$\begin{aligned}\vec{A} \cdot \vec{B} &= A_x B_x + A_y B_y + A_z B_z \\ &= (2.00)(-4.00) + (3.00)(2.00) + (1.00)(-1.00) \\ &= -3.00 \\ A &= \sqrt{A_x^2 + A_y^2 + A_z^2} = \sqrt{(2.00)^2 + (3.00)^2 + (1.00)^2} \\ &= \sqrt{14.00} \\ B &= \sqrt{B_x^2 + B_y^2 + B_z^2} = \sqrt{(-4.00)^2 + (2.00)^2 + (-1.00)^2} \\ &= \sqrt{21.00} \\ \cos \phi &= \frac{A_x B_x + A_y B_y + A_z B_z}{AB} = \frac{-3.00}{\sqrt{14.00} \sqrt{21.00}} = -0.175 \\ \phi &= 100^\circ\end{aligned}$$

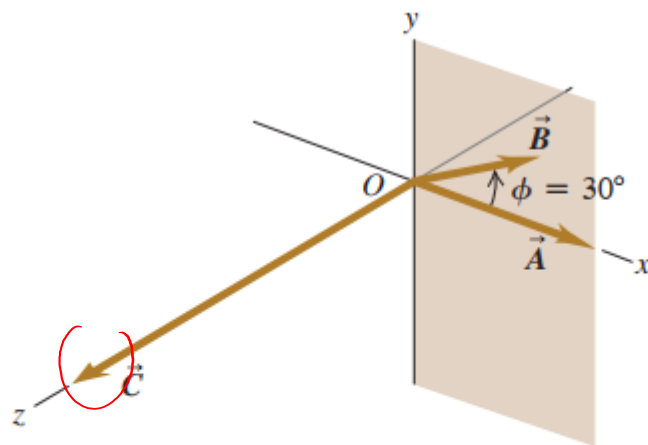
EVALUATE: As a check on this result, note that the scalar product $\vec{A} \cdot \vec{B}$ is negative. This means that ϕ is between 90° and 180° (see Fig. 1.26), which agrees with our answer.

EXAMPLE 1.11 Calculating a vector product

Vector $\vec{A}$ has magnitude 6 units and is in the direction of the $+x$ -axis. Vector $\vec{B}$ has magnitude 4 units and lies in the xy -plane, making an angle of 30° with the $+x$ -axis (**Fig. 1.33**). Find the vector product $\vec{C} = \vec{A} \times \vec{B}$.

IDENTIFY and SET UP We'll find the vector product in two ways, which will provide a check of our calculations. First we'll use Eq. (1.20) and the right-hand rule; then we'll use Eqs. (1.25) to find the vector product by using components.

Figure 1.33 Vectors $\vec{A}$ and $\vec{B}$ and their vector product $\vec{C} = \vec{A} \times \vec{B}$. Vector $\vec{B}$ lies in the xy -plane.



EXECUTE From Eq. (1.20) the magnitude of the vector product is

$$AB \sin \phi = (6)(4)(\sin 30^\circ) = 12$$

By the right-hand rule, the direction of $\vec{A} \times \vec{B}$ is along the $+z$ -axis (the direction of the unit vector $\hat{k}$), so $\vec{C} = \vec{A} \times \vec{B} = 12\hat{k}$.

To use Eqs. (1.25), we first determine the components of $\vec{A}$ and $\vec{B}$. Note that $\vec{A}$ points along the x -axis, so its only nonzero component is A_x . For $\vec{B}$, Fig. 1.33 shows that $\phi = 30^\circ$ is measured from the $+x$ -axis toward the $+y$ -axis, so we can use Eqs. (1.5):

$$\begin{array}{lll} A_x = 6 & A_y = 0 & A_z = 0 \\ B_x = 4 \cos 30^\circ = 2\sqrt{3} & B_y = 4 \sin 30^\circ = 2 & B_z = 0 \end{array}$$

Then Eqs. (1.25) yield

$$\begin{array}{l} C_x = (0)(0) - (0)(2) = 0 \\ C_y = (0)(2\sqrt{3}) - (6)(0) = 0 \\ C_z = (6)(2) - (0)(2\sqrt{3}) = 12 \end{array}$$

Thus again we have $\vec{C} = 12\hat{k}$.

EVALUATE Both methods give the same result. Depending on the situation, one or the other of the two approaches may be the more convenient one to use.

KEYCONCEPT The vector product $\vec{A} \times \vec{B}$ of two vectors is a third vector that is perpendicular to both $\vec{A}$ and $\vec{B}$. You can find the vector product either from the magnitudes of the two vectors, the angle between them, and the right-hand rule, or from the components of the two vectors.

A robotic vehicle, or rover, is exploring the surface of Mars. The stationary Mars lander is the origin of coordinates, and the surrounding Martian surface lies in the xy -plane. The rover, which we represent as a point, has x - and y -coordinates that vary with time:

$$x = 2.0 \text{ m} - (0.25 \text{ m/s}^2)t^2$$

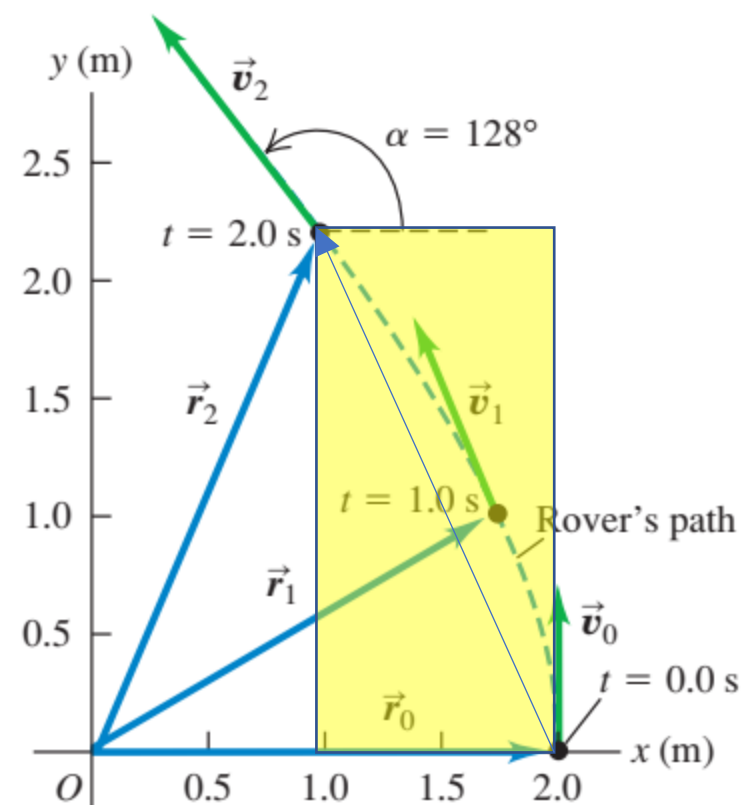
$$y = (1.0 \text{ m/s})t + (0.025 \text{ m/s}^3)t^3$$

(a) Find the rover's coordinates and distance from the lander at $t = 2.0 \text{ s}$. (b) Find the rover's displacement and average velocity vectors for the interval $t = 0.0 \text{ s}$ to $t = 2.0 \text{ s}$. (c) Find a general expression for the rover's instantaneous velocity vector $\vec{v}$. Express $\vec{v}$ at $t = 2.0 \text{ s}$ in component form and in terms of magnitude and direction.

SOLUTION

IDENTIFY and SET UP: This problem involves motion in two dimensions, so we must use the vector equations obtained in this section. Figure 3.5 shows the rover's path (dashed line). We'll use Eq. (3.1) for position $\vec{r}$, the expression $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$ for displacement, Eq. (3.2) for average velocity, and Eqs. (3.5), (3.6), and (3.7)

3.5 At $t = 0.0 \text{ s}$ the rover has position vector $\vec{r}_0$ and instantaneous velocity vector $\vec{v}_0$. Likewise, $\vec{r}_1$ and $\vec{v}_1$ are the vectors at $t = 1.0 \text{ s}$; $\vec{r}_2$ and $\vec{v}_2$ are the vectors at $t = 2.0 \text{ s}$.



for instantaneous velocity and its magnitude and direction. The target variables are stated in the problem.

EXECUTE: (a) At $t = 2.0$ s the rover's coordinates are

$$x = 2.0 \text{ m} - (0.25 \text{ m/s}^2)(2.0 \text{ s})^2 = 1.0 \text{ m}$$

$$y = (1.0 \text{ m/s})(2.0 \text{ s}) + (0.025 \text{ m/s}^3)(2.0 \text{ s})^3 = 2.2 \text{ m}$$

The rover's distance from the origin at this time is

$$r = \sqrt{x^2 + y^2} = \sqrt{(1.0 \text{ m})^2 + (2.2 \text{ m})^2} = 2.4 \text{ m}$$

(b) To find the displacement and average velocity over the given time interval, we first express the position vector $\vec{r}$ as a function of time t . From Eq. (3.1) this is

$$\begin{aligned}\vec{r} &= x\hat{i} + y\hat{j} \\ &= [2.0 \text{ m} - (0.25 \text{ m/s}^2)t^2]\hat{i} \\ &\quad + [(1.0 \text{ m/s})t + (0.025 \text{ m/s}^3)t^3]\hat{j}\end{aligned}$$

At $t = 0.0$ s the position vector $\vec{r}_0$ is

$$\vec{r}_0 = (2.0 \text{ m})\hat{i} + (0.0 \text{ m})\hat{j}$$

From part (a), the position vector $\vec{r}_2$ at $t = 2.0$ s is

$$\vec{r}_2 = (1.0 \text{ m})\hat{i} + (2.2 \text{ m})\hat{j}$$

The displacement from $t = 0.0$ s to $t = 2.0$ s is therefore

$$\begin{aligned}\Delta\vec{r} &= \vec{r}_2 - \vec{r}_0 = (1.0 \text{ m})\hat{i} + (2.2 \text{ m})\hat{j} - (2.0 \text{ m})\hat{i} \\ &= (-1.0 \text{ m})\hat{i} + (2.2 \text{ m})\hat{j}\end{aligned}$$

During this interval the rover moves 1.0 m in the negative x -direction and 2.2 m in the positive y -direction. From Eq. (3.2), the average velocity over this interval is the displacement divided by the elapsed time:

$$\begin{aligned}\vec{v}_{\text{av}} &= \frac{\Delta\vec{r}}{\Delta t} = \frac{(-1.0 \text{ m})\hat{i} + (2.2 \text{ m})\hat{j}}{2.0 \text{ s} - 0.0 \text{ s}} \\ &= (-0.50 \text{ m/s})\hat{i} + (1.1 \text{ m/s})\hat{j}\end{aligned}$$

The components of this average velocity are $v_{\text{av-}x} = -0.50 \text{ m/s}$ and $v_{\text{av-}y} = 1.1 \text{ m/s}$.

(c) From Eq. (3.4) the components of *instantaneous* velocity are the time derivatives of the coordinates:

$$v_x = \frac{dx}{dt} = (-0.25 \text{ m/s}^2)(2t)$$

$$v_y = \frac{dy}{dt} = 1.0 \text{ m/s} + (0.025 \text{ m/s}^3)(3t^2)$$

Hence the instantaneous velocity vector is

$$\vec{v} = v_x \hat{i} + v_y \hat{j} = (-0.50 \text{ m/s}^2)t \hat{i} + [1.0 \text{ m/s} + (0.075 \text{ m/s}^3)t^2] \hat{j}$$

At $t = 2.0 \text{ s}$ the velocity vector $\vec{v}_2$ has components

$$v_{2x} = (-0.50 \text{ m/s}^2)(2.0 \text{ s}) = -1.0 \text{ m/s}$$

$$v_{2y} = 1.0 \text{ m/s} + (0.075 \text{ m/s}^3)(2.0 \text{ s})^2 = 1.3 \text{ m/s}$$

The magnitude of the instantaneous velocity (that is, the speed) at $t = 2.0 \text{ s}$ is

$$v_2 = \sqrt{v_{2x}^2 + v_{2y}^2} = \sqrt{(-1.0 \text{ m/s})^2 + (1.3 \text{ m/s})^2}$$

$$= 1.6 \text{ m/s}$$

Figure 3.5 shows the direction of the velocity vector $\vec{v}_2$, which is at an angle α between 90° and 180° with respect to the positive x -axis. From Eq. (3.7) we have

$$\arctan \frac{v_y}{v_x} = \arctan \frac{1.3 \text{ m/s}}{-1.0 \text{ m/s}} = -52^\circ$$

This is off by 180° ; the correct value of the angle is $\alpha = 180^\circ - 52^\circ = 128^\circ$, or 38° west of north.

EVALUATE: Compare the components of *average* velocity that we found in part (b) for the interval from $t = 0.0 \text{ s}$ to $t = 2.0 \text{ s}$ ($v_{\text{av-}x} = -0.50 \text{ m/s}$, $v_{\text{av-}y} = 1.1 \text{ m/s}$) with the components of *instantaneous* velocity at $t = 2.0 \text{ s}$ that we found in part (c) ($v_{2x} = -1.0 \text{ m/s}$, $v_{2y} = 1.3 \text{ m/s}$). The comparison shows that, just as in one dimension, the average velocity vector $\vec{v}_{\text{av}}$ over an interval is in general *not* equal to the instantaneous velocity $\vec{v}$ at the end of the interval (see Example 2.1).

Figure 3.5 shows the position vectors $\vec{r}$ and instantaneous velocity vectors $\vec{v}$ at $t = 0.0 \text{ s}$, 1.0 s , and 2.0 s . (You should calculate these quantities for $t = 0.0 \text{ s}$ and $t = 1.0 \text{ s}$.) Notice that $\vec{v}$ is tangent to the path at every point. The magnitude of $\vec{v}$ increases as the rover moves, which means that its speed is increasing.